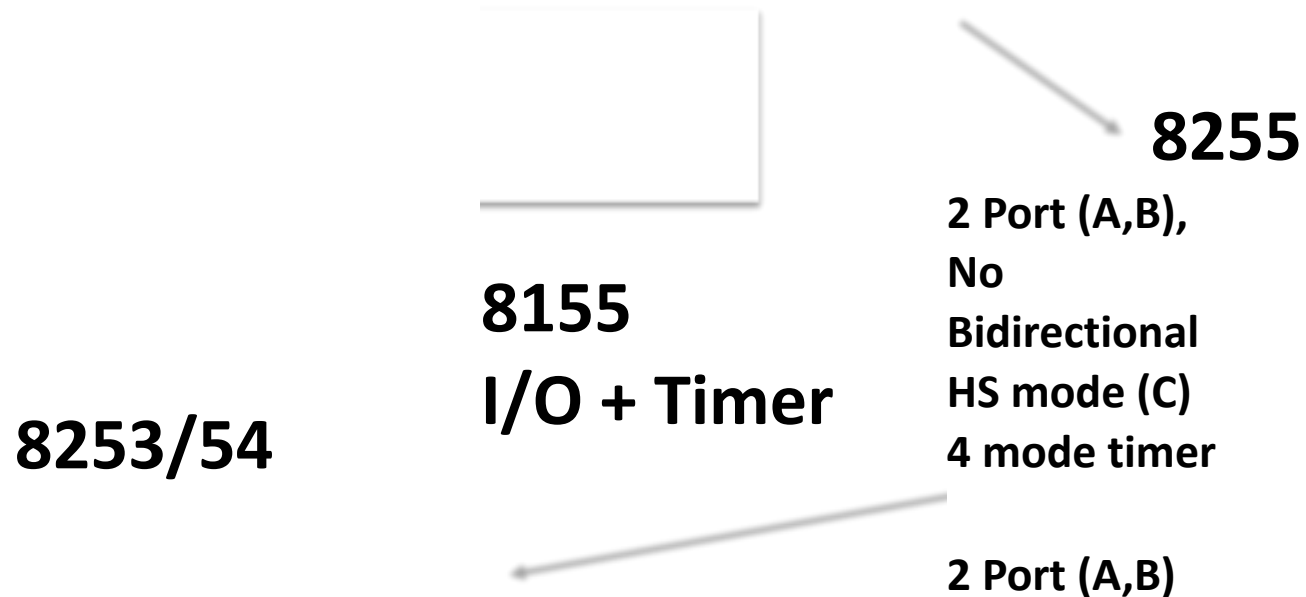


Dedicated Peripheral Interface Device (Introduction to 8255)

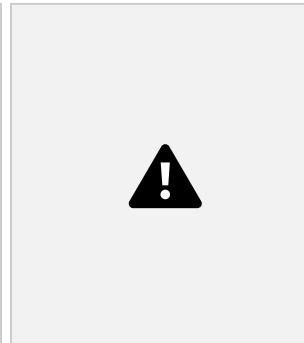
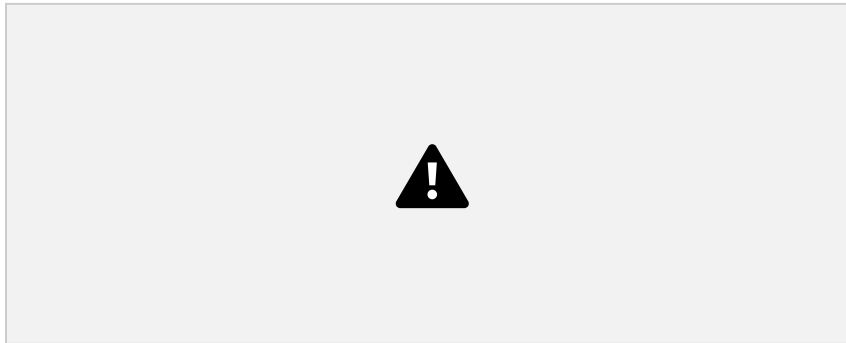
Hierarchy of I/O Control Devices





Timer 8259

6 mode timer



8237



I/O

A is Bidirectional HS mode (C)

Extra controls **8251**



DMA controller

**Serial I/O USART
controller**

Interrupt controller

- **8155 I/O Interface & Timer**



– Dedicated I/O interface (8255)

– Dedicated Timer (8254/8253)

• 8255 Ports and mode of operations •

Interfacing A/D Converter using Handshake mode using

• IO e 6

Capability: bit

– 2kbits I/O port •

static RAM Timer

256x8

– 2

programmabl

e 8 bit I/O

ports – 1

programmabl

– 1

programmabl

e 14 bit

binary

counter/time

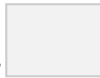
– 4 Modes

Reset in RD
WR

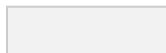
C
IO/M_E

AD0-AD7
E

Timer
CLK



AL

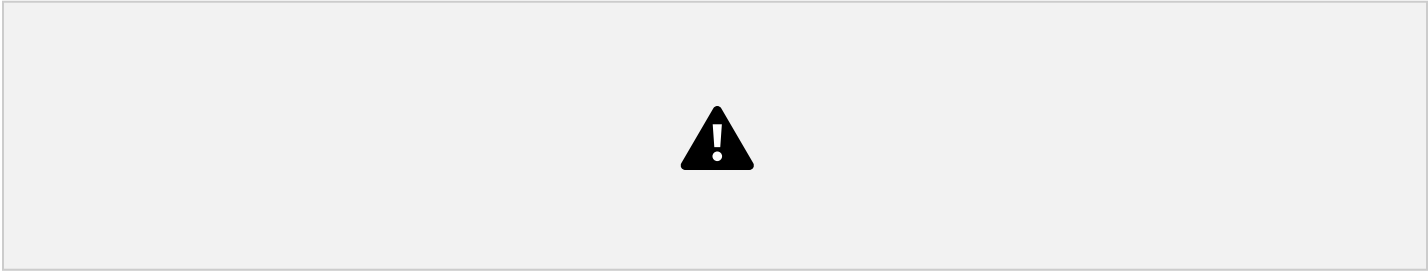




N ☐ ☐ ☐

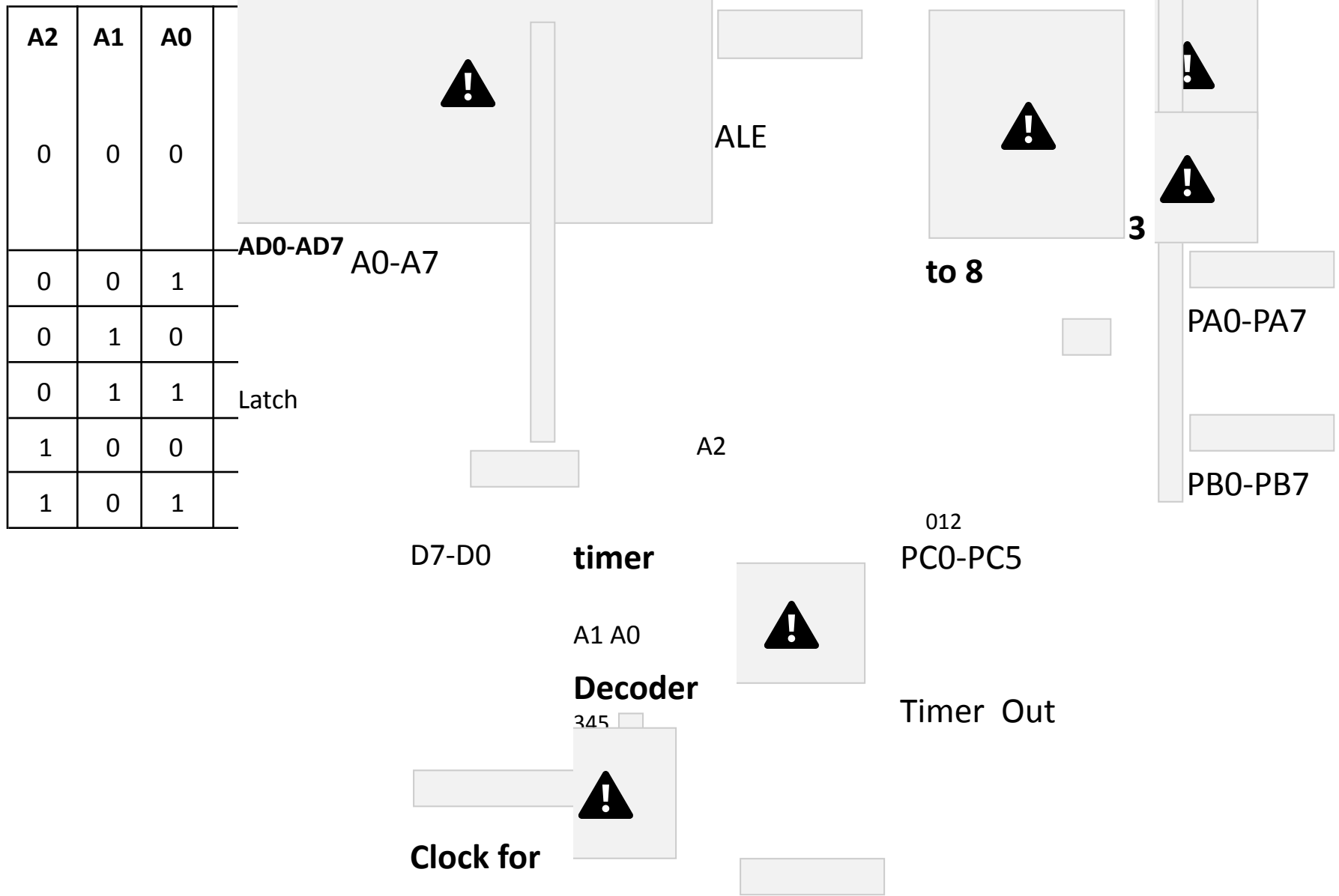
A L T	D 3	D 2	F
1	0	0	
2	0	1	C
3	1	0	C
4	1	1	S

N



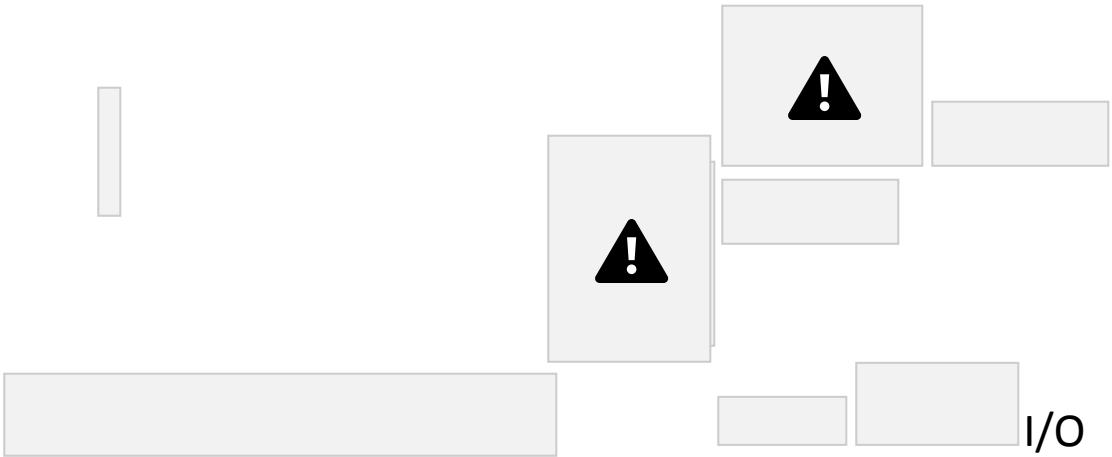
CE^b





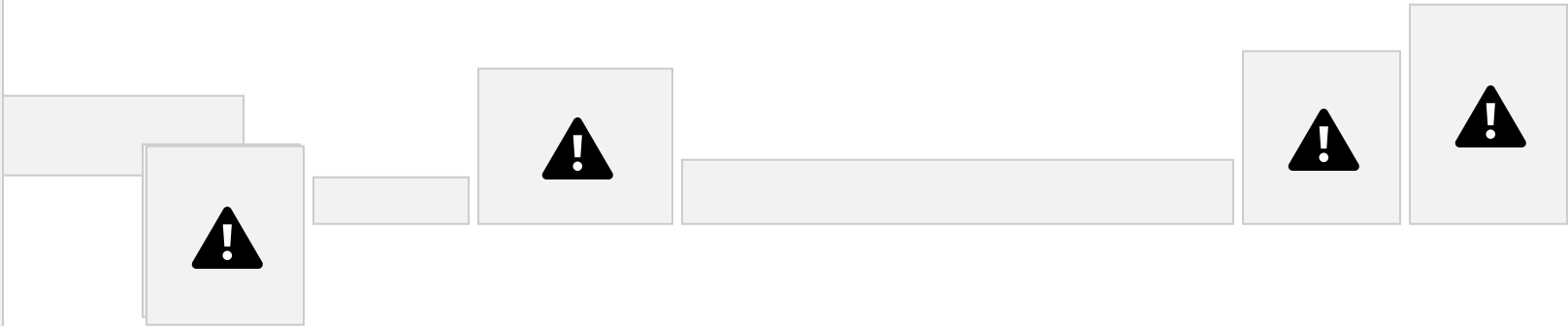
Block Diagram of 8255

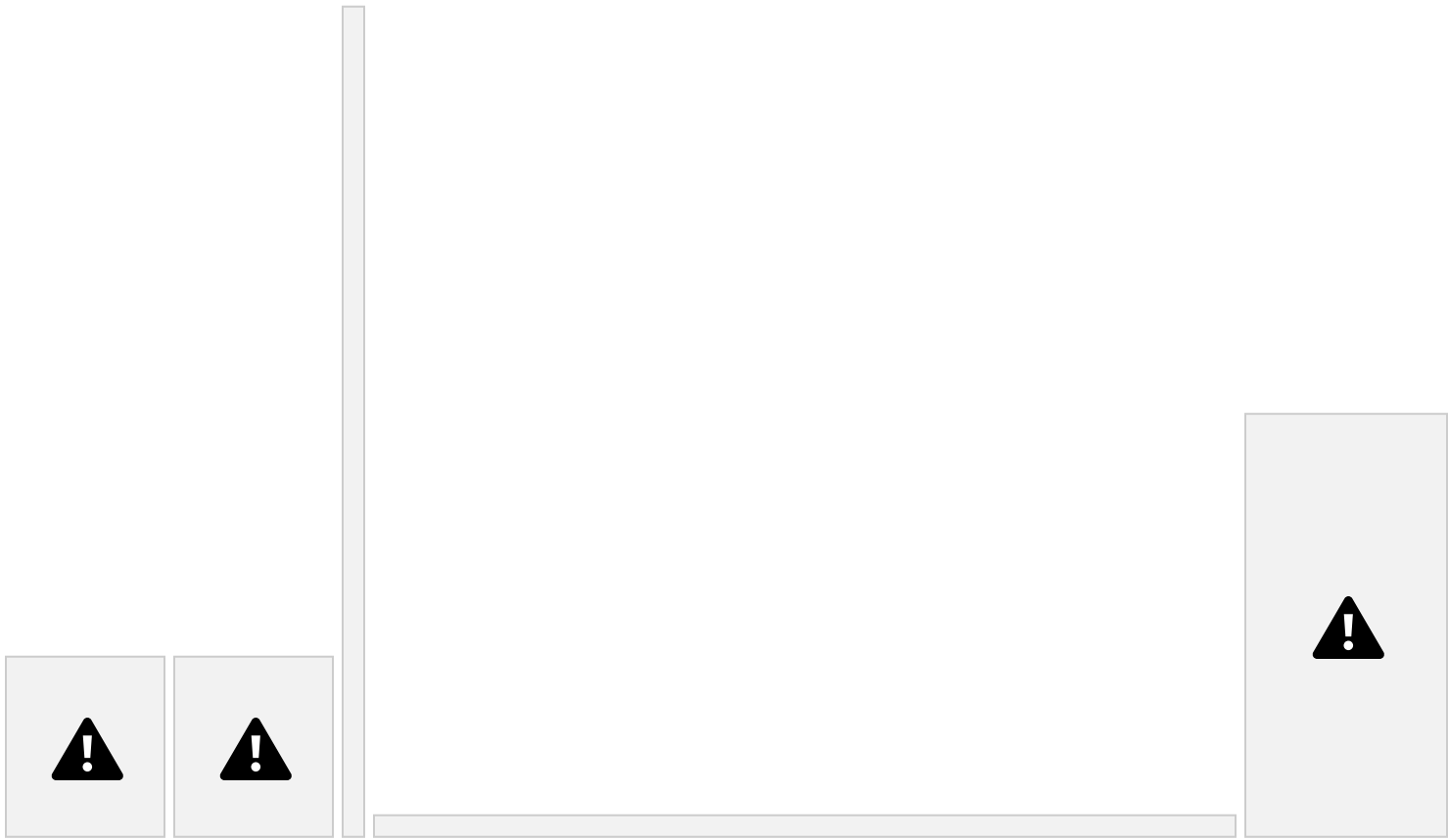
0	0	0	Pc
0	0	1	Pc
0	1	0	Pc
0	1	1	C



PA7-PA0

Bi directional





I/O

Data Bus D7-D0

8 bit Internal
Data Bus

PC7-PC4

I/O
PC3-PC0

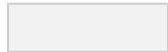
RD^b WR^b

I/O

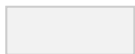
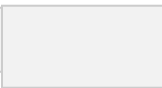
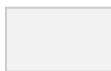
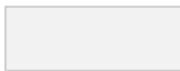
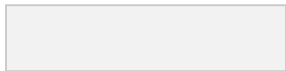
A1

A0

RESET



CS^b



PB7-PB0

in 8255

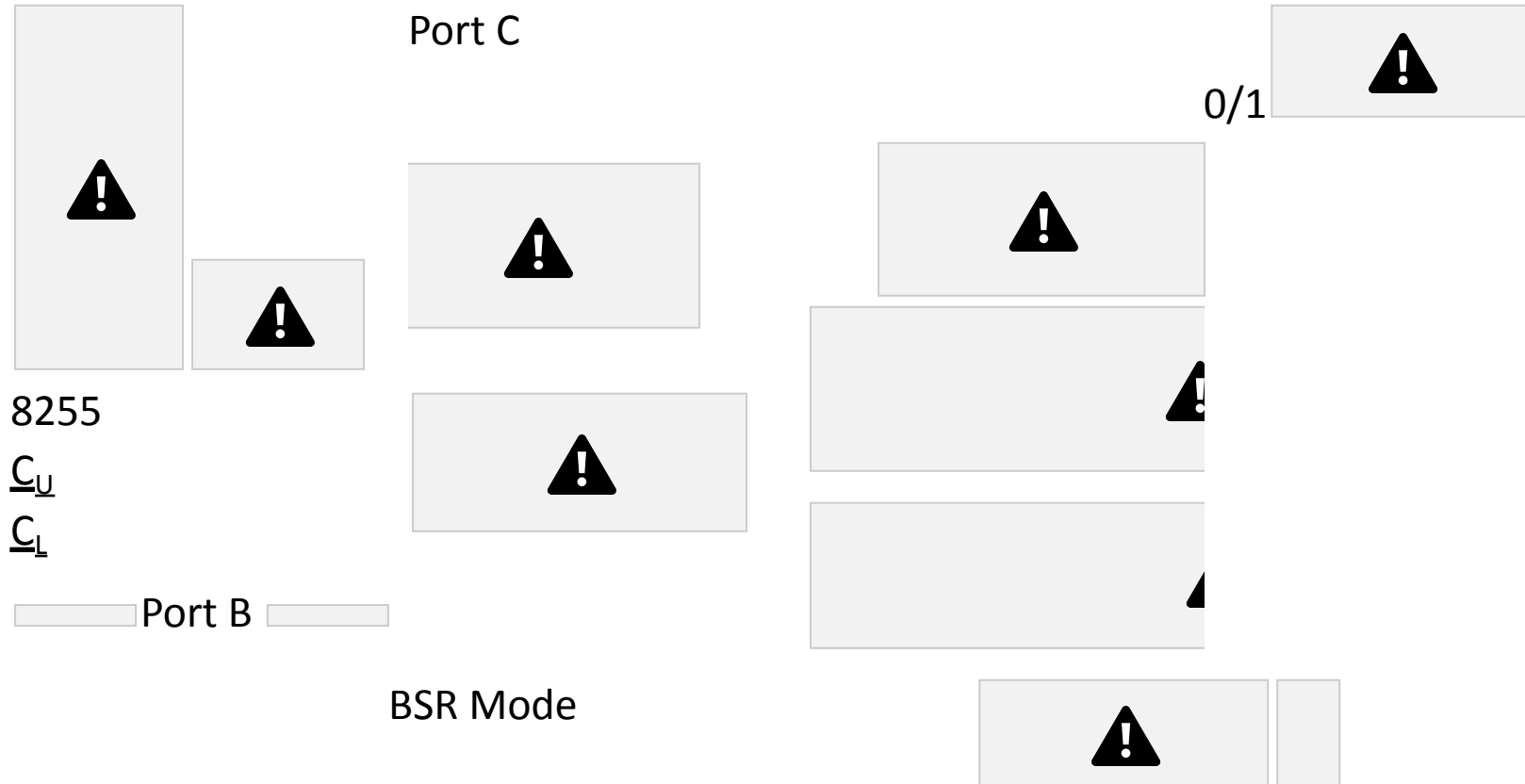
D7 D6 D5 D4

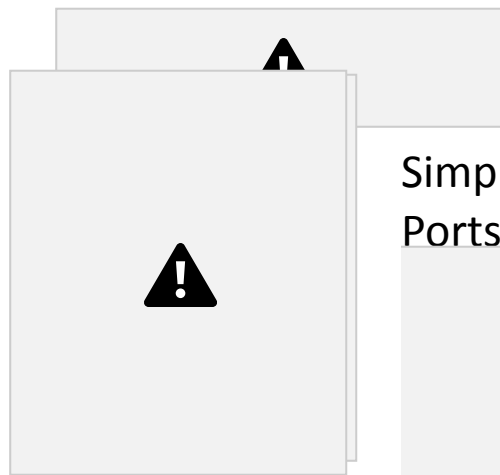
Port A

Ports & Modes^{D3 D2 D1 D}

Port C

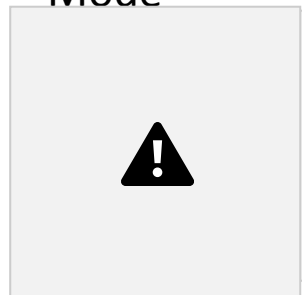
0/1





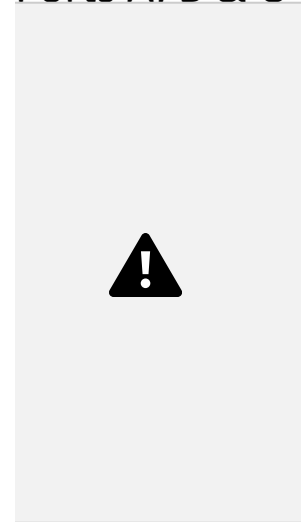
Bit Set/Reset

For Port C
No Effect on I/O
Mode



Mode 0

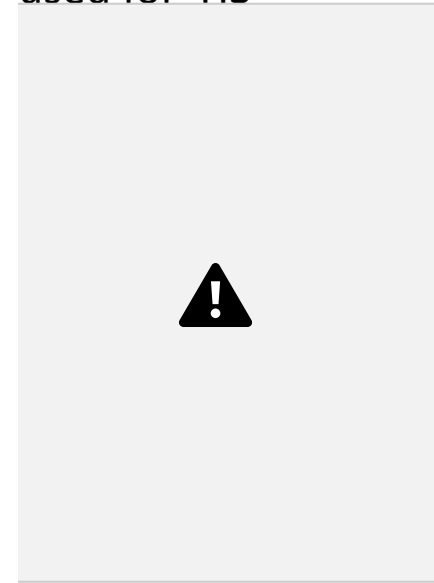
Simple I/O for
Ports A, B & C



Mode 1
HS mode
for Ports
A and/or B

Port C bits are

used for HS



Mode 2
Bidirectional

Data mode for
Port A

B can in mode
0/1

Port C bits are
used for HS

Ports & Modes in 8255 : Control register



7 6 5 4 3 2 1 0

D7 D6 D5 D4 D3 D2 D1 D0

Group B

Port C(L) – 1 Input
0 output

Port B – 1 Input 0 output

Mode select: 0 mode 0; 1 mode 1

Port C(U) – 1 Input 0 output

Port A – 1 Input 0 output

Mode select: 00 mode 0;
01 mode 1; 0x

Group A

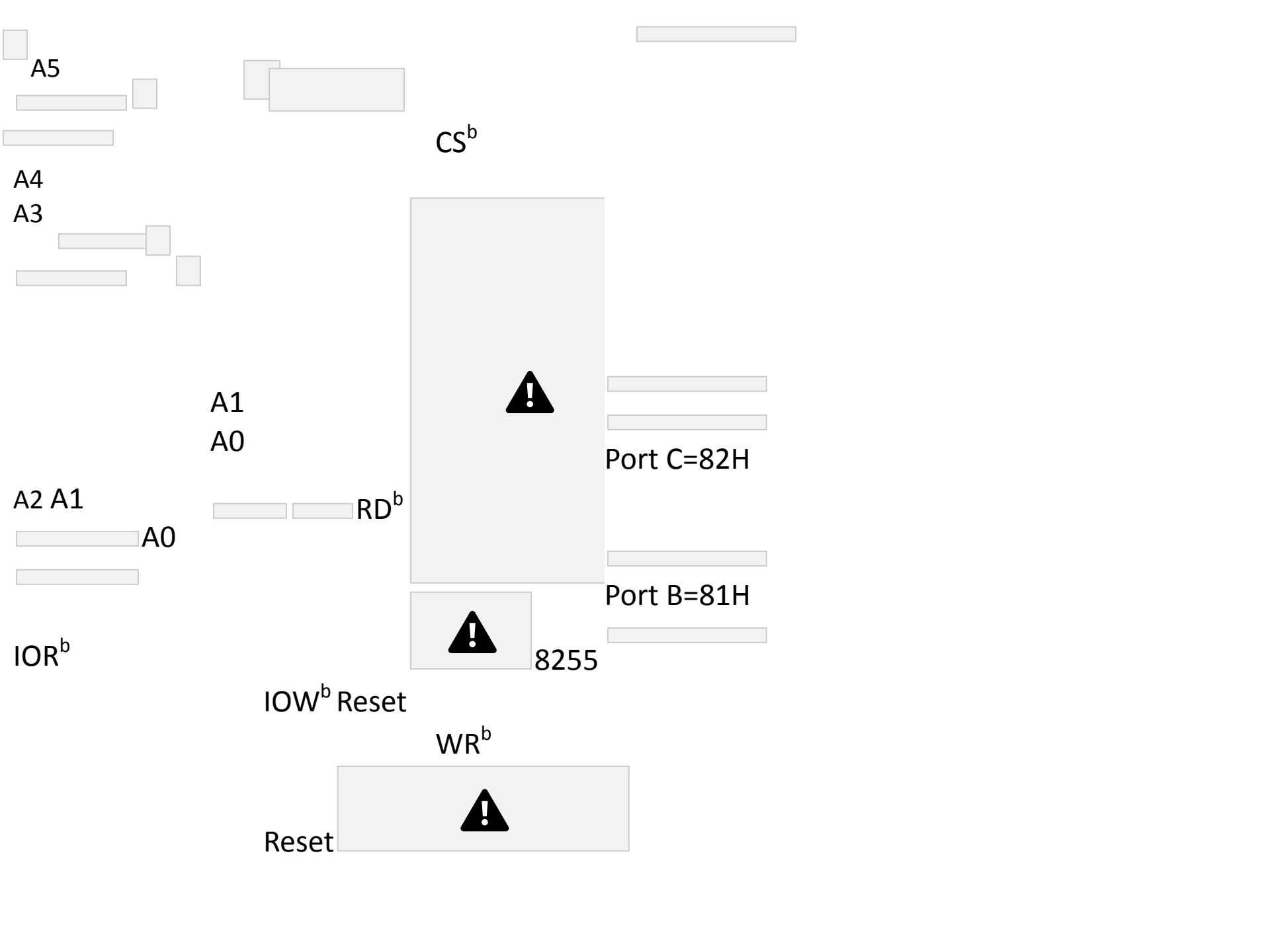
1 – mode select 0 – mode 2
bit set/reset

I/O port Addressing

A7

Port A=80H

A6



CS^b	A1 A0	HEX Address	Port
A7 A6 A5 A4 A3 A2 1 0 0 0 0 0	A1 A0 0 0	= 80H	A
	0 1	=81H	B
	1 0	=82H	C
	1 1	=83H	Control Register

BSR (Bit Set or Reset Mode)

- Set/Reset bit of Port C
- Heavily used for HS and Interrupt mode
- BSR Control word

D7 0 BSR Mode	D6 D5 D4 Not used, So (000)	D3 D2 D1 Bit Select	D0 S/R (1/0)
-------------------------------	---------------------------------------	-------------------------------	------------------------

- BSR Control word
 - To set PC7= 0 000 111 1 (0FH)
 - To reset PC7= 0 000 111 0 (0EH)
 - To set PC3 = 0 000 011 1 (07H)

Ports

- Control register controls the overall operation of 8255
- All three ports A, B and C are grouped into two

Group B

Group A



Port A Upper
C Lower C Port
B

Operation modes

- 8255 has three modes:
 - mode 0: basic input-output
 - mode 1: strobed input-output
 - mode 2: strobed bidirectional bus

I/O • In mode 0

- two 8-bit ports and two 4-bit ports - any port can be input or output

- Outputs are latched, inputs are not latched
- ## Mode 0



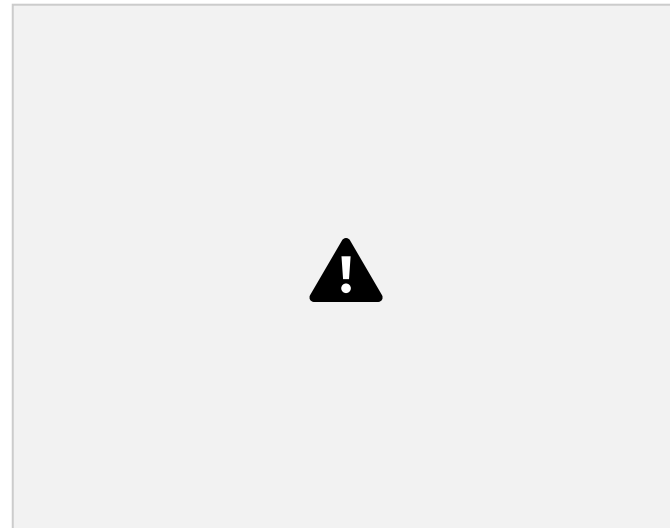
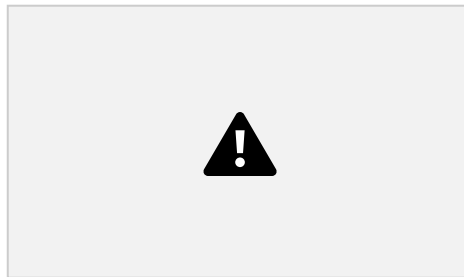
Port A Upper C Lower C Port B

Operation mode 1

- In mode 1:
 - three ports are divided into two groups
 - each group contains one 8-bit port and one 4-bit control/data port
 - 8-bit port can be either input or output

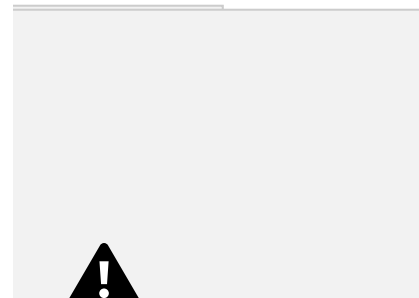
and both latched

- 4-bit port used for control and status of 8-bit data port

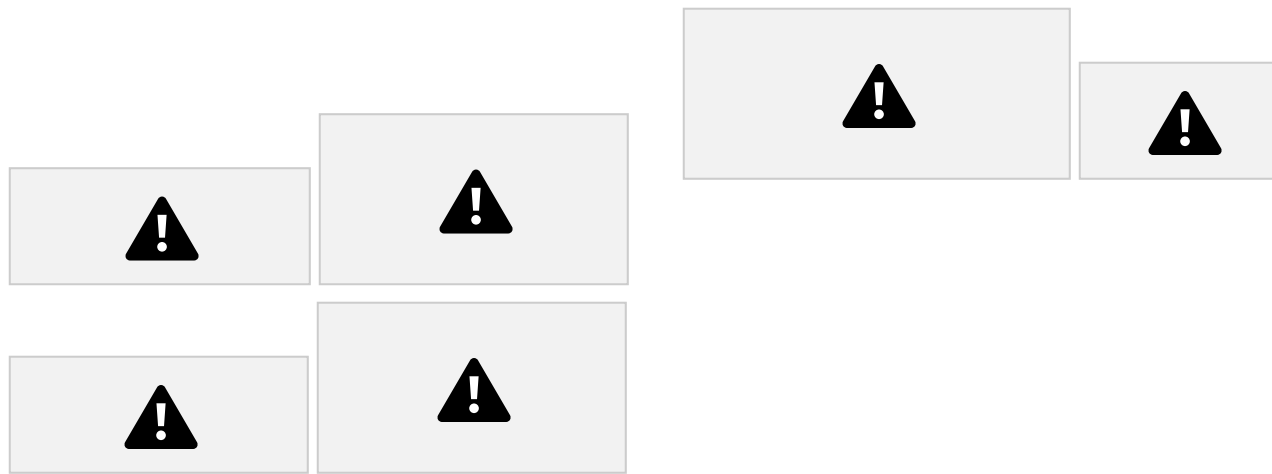


Group A

Mode 1



Group B



Port A Upper C Lower C Port B

Operation mode 2

- In mode 2
 - only port A is used
 - port A becomes an 8-bit bidirectional bus - port C acts as control port (only pins PC3-PC7 are used)

Mode 2



Group A

Group B

Port A C7-C3 Port B

Programming 8255

☐ Mode 2:

- Port A is programmed to be bi-directional
- Port C is for handshaking
- Port B can be either input or output in mode 0 or mode

1 PA[7:0]

PC7

PC6

PC4

8255 PC5 PC3

PC0

PC0

PC0

OBF_A

ACK_A

STB_A

IBF_A

INTR_A

Mode 0

ACK_B $INTR_B$

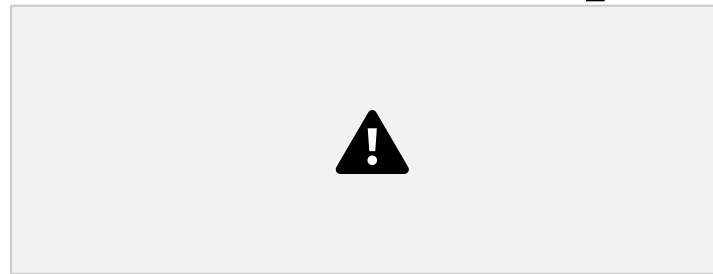
$INTR_B$ Mode 1

PB[7:0]

In Out In Out

In Out

STB_B OBF_B IBF_B



- R S Gaonkar, “Microprocessor Architecture”, Chapter 15